9.1 Sample Spaces and Probability

Name _____ Period ____ Date _____

Algebra I 9.1 · Student sheet

I can…

- list the outcomes in a sample space and count them.
- find theoretical probabilities and the probability of a complement.
- find geometric and experimental probabilities.

Vocabulary · fill in the missing word

Probability experiment — an action or trial with _____ results

Outcome — one possible result of an _____

Event — a _____ of one or more outcomes

Sample space — the set of all _____ outcomes

Probability — a number from 0 (_____) to 1 (certain) that measures how likely an event is

Theoretical probability — (number of _____ outcomes) ÷ (total number of outcomes), when outcomes are equally likely

Complement of A — all _____ not in A; P(not A) = 1 − P(A)

Geometric probability — a probability found from _____ or areas

Experimental probability — (number of _____) ÷ (number of trials)

Part 1 · Coin and die

List the sample space for flipping a coin and rolling a die.

	1	2	3	4	5	6
Heads						
Tails						

Flip and roll 20 times. Tally how often you get **heads and an even number**.

Tally	Successes	Trials	Experimental P	Theoretical P
		20		

Class total: _____ successes out of _____ trials = _____. Is it closer to the theoretical probability?

Part 2 · Notes: fill in the blanks

Theoretical: P(A) = _____ ÷ _____ **Complement:** P(not A) = _____

Geometric: P = (favorable _____ or area) ÷ (total length or _____)

Experimental: P = _____ ÷ _____

Probability 0 means _____; probability 1 means _____.

Part 3 · Examples (work them with the class)

Example 1

You flip a coin and spin a spinner with three equal sections: red, blue, and green. List the sample space. How many outcomes are there?

Example 2

You guess on three true-false questions. Find the probability that you get

a. all three correct. b. exactly two correct.

Example 3

Two six-sided dice are rolled. What is the probability that the sum is not 7?

Example 4

A dart lands at a random point on a square board 20 inches on a side. A circle with a radius of 5 inches is in the center. What is the probability that the dart lands in the circle?

Example 5

A die is rolled 60 times. The table shows the results. What is the experimental probability of rolling a 2? How does it compare with the theoretical probability?

Number	1	2	3	4	5	6
Times rolled	8	12	9	11	10	10

Part 4 · Partner practice

Partner 1

A spinner has 8 equal sections numbered 1 to 8. Find P(a number greater than 5) and P(a number not greater than 5).

Partner 2

A coin lands at a random point on a 10-inch by 8-inch board. A 4-inch by 2-inch rectangle is painted on the board. What is the probability that the coin lands in the painted rectangle?